ISLR

Kam

```
#install.packages("Hmisc", "rms", "MASS", "glmnet", "glmnetUtils", "pls", "leaps")
#install.packages("glmnetUtils")
#install.packages("pls")

library(Hmisc)
```

```
Warning: package 'Hmisc' was built under R version 4.4.3
```

```
Attaching package: 'Hmisc'
```

```
The following objects are masked from 'package:base':

    format.pval, units
```

```
library(rms)
```

```
Warning: package 'rms' was built under R version 4.4.3
```

```
library(MASS)
library(glmnet)
```

```
Warning: package 'glmnet' was built under R version 4.4.3
```

```
Loading required package: Matrix
```

```
Loaded glmnet 4.1-10
```

```r
library(glmnetUtils)
```

```
Warning: package 'glmnetUtils' was built under R version 4.4.3
```

```
Attaching package: 'glmnetUtils'
```

```
The following objects are masked from 'package:glmnet':

    cv.glmnet, glmnet
```

```r
library(pls) # subset selection and pca
```

```
Warning: package 'pls' was built under R version 4.4.3
```

```
Attaching package: 'pls'
```

```
The following object is masked from 'package:stats':

    loadings
```

```r
set.seed(2026)
d <- MASS::birthwt
stopifnot(nrow(d)==189, "bwt" %in% names(d))
str(d)
```

```
'data.frame':   189 obs. of  10 variables:
 $ low  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ age  : int  19 33 20 21 18 21 22 17 29 26 ...
 $ lwt  : int  182 155 105 108 107 124 118 103 123 113 ...
 $ race : int  2 3 1 1 1 3 1 3 1 1 ...
 $ smoke: int  0 0 1 1 1 0 0 0 1 1 ...
 $ ptl  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ ht   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ ui   : int  1 0 0 1 1 0 0 0 0 0 ...
 $ ftv  : int  0 3 1 2 0 0 1 1 1 0 ...
 $ bwt  : int  2523 2551 2557 2594 2600 2622 2637 2637 2663 2665 ...
```

```r
library(MASS)

# Restore original data
d <- MASS::birthwt

# Remove outcome-derived variable
d$low <- NULL

# Recode categorical predictors
d$race <- factor(d$race, levels = c(1, 2, 3),
                 labels = c("White", "Black", "Other"))

d$smoke <- factor(d$smoke, levels = c(0, 1),
                  labels = c("No", "Yes"))

d$ht <- factor(d$ht, levels = c(0, 1),
               labels = c("No", "Yes"))

d$ui <- factor(d$ui, levels = c(0, 1),
               labels = c("No", "Yes"))

d$ptl <- factor(d$ptl > 0,
                levels = c(FALSE, TRUE),
                labels = c("No", "Yes"))

d$ftv_cat <- factor(
  ifelse(d$ftv == 0, "0",
         ifelse(d$ftv == 1, "1", "2+")),
  levels = c("0", "1", "2+")
)

d$ftv <- NULL

# Validate before plotting
stopifnot(
  nrow(d) == 189,
  all(!is.na(d$smoke)),
  all(!is.na(d$ptl)),
  is.numeric(d$bwt)
)

print(table(d$smoke, useNA = "ifany"))
```

```
 No Yes
115  74
```

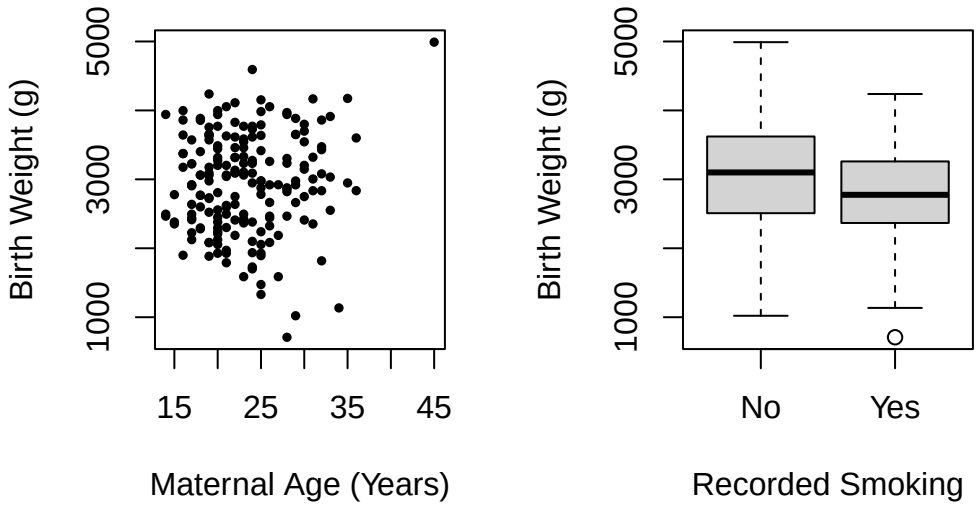
```r
print(summary(d$bwt))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    709    2414    2977    2945    3487    4990
```

```r
# Plot
par(mfrow = c(1, 2))

plot(bwt ~ age, data = d,
     pch = 16, cex = 0.65,
     xlab = "Maternal Age (Years)",
     ylab = "Birth Weight (g)")

boxplot(bwt ~ smoke, data = d,
        xlab = "Recorded Smoking",
        ylab = "Birth Weight (g)")
```



```r
par(mfrow = c(1, 1))
```

```r
library(rms)

dd <- datadist(d)
options(datadist = "dd")

fit1 <- ols(bwt ~ lwt,
            data = d,
            x = TRUE,
            y = TRUE)

print(fit1)
```

```
Linear Regression Model

ols(formula = bwt ~ lwt, data = d, x = TRUE, y = TRUE)

                 Model Likelihood    Discrimination
                       Ratio Test           Indexes
Obs       189    LR chi2      6.64    R2       0.034
sigma718.4394    d.f.            1    R2 adj   0.029
d.f.      187    Pr(> chi2) 0.0100    g      142.262

Residuals

     Min       1Q   Median       3Q      Max
-2192.12  -497.97    -3.84   508.32  2075.60

          Coef      S.E.     t     Pr(>|t|)
Intercept 2369.6235 228.4932 10.37 <0.0001
lwt          4.4291   1.7135  2.58 0.0105
```

```r
# multiple regression

dd_multiple <- rms::datadist(d)
options(datadist="dd")
fit_multiple <- rms::ols(bwt ~ age + lwt + race + smoke + ptl, data=d, x=TRUE, y=TRUE)
print(fit_multiple)
```

```
Linear Regression Model
```

```
rms::ols(formula = bwt ~ age + lwt + race + smoke + ptl, data = d,
    x = TRUE, y = TRUE)

                 Model Likelihood    Discrimination
                       Ratio Test           Indexes
Obs       189    LR chi2     34.95    R2       0.169
sigma675.6812    d.f.            6    R2 adj   0.141
d.f.      182    Pr(> chi2) 0.0000    g      336.138

Residuals

      Min        1Q    Median        3Q       Max
-2313.614  -395.080    -8.108   514.283  1660.275

           Coef      S.E.     t     Pr(>|t|)
Intercept  2834.3399 318.4157  8.90 <0.0001
age           1.2555   9.8442  0.13 0.8987
lwt           3.5682   1.7336  2.06 0.0410
race=Black -484.9353 156.0636 -3.11 0.0022
race=Other -369.4043 119.2526 -3.10 0.0023
smoke=Yes  -350.7989 110.8440 -3.16 0.0018
ptl=Yes    -296.8466 139.9815 -2.12 0.0353
```

```
coef(fit_multiple)
```

```
  Intercept         age         lwt  race=Black  race=Other   smoke=Yes
2834.339900    1.255530    3.568184 -484.935281 -369.404336 -350.798927
    ptl=Yes
-296.846629
```

```
anova(fit_multiple)
```

```
                Analysis of Variance          Response: bwt

 Factor     d.f. Partial SS   MS          F     P
 age           1      7426.376    7426.376  0.02 0.8987
 lwt           1   1934011.119 1934011.119  4.24 0.0410
 race          2   6755126.332 3377563.166  7.40 0.0008
 smoke         1   4572735.498 4572735.498 10.02 0.0018
 ptl           1   2053083.370 2053083.370  4.50 0.0353
```

```
REGRESSION   6  16878454.526 2813075.754  6.16 <.0001
ERROR      182  83091201.284  456545.062
```

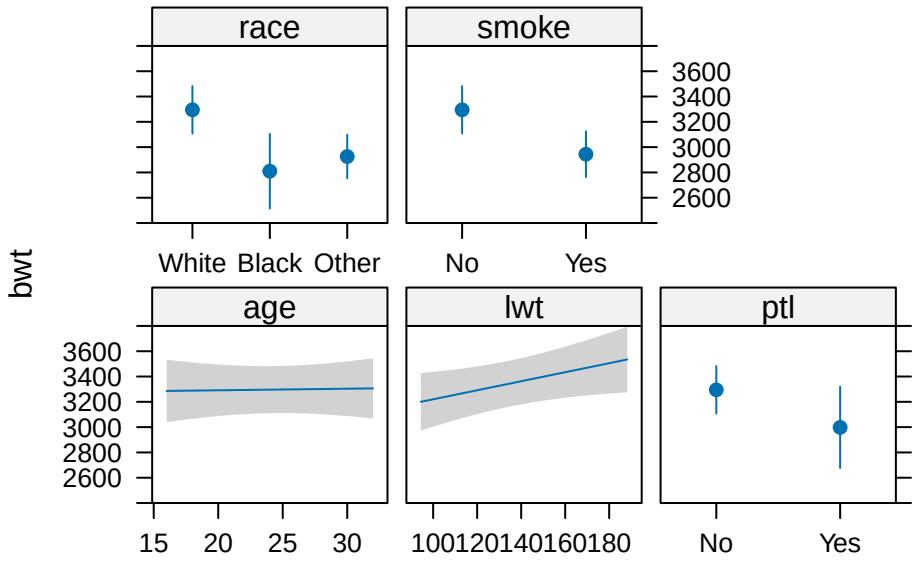
```
summary(fit_multiple)
```

```
            Effects              Response : bwt

Factor              Low High Diff. Effect    S.E.    Lower 0.95 Upper 0.95
age                  19   26   7       8.7887  68.909 -127.1800   144.750
lwt                 110  140   30     107.0500  52.009    4.4268   209.660
race - Black:White    1    2   NA    -484.9400 156.060 -792.8600  -177.010
race - Other:White    1    3   NA    -369.4000 119.250 -604.7000  -134.110
smoke - Yes:No        1    2   NA    -350.8000 110.840 -569.5000  -132.090
ptl - Yes:No          1    2   NA    -296.8500 139.980 -573.0400   -20.651
```

Use rms to predict model conditional mean

```
plot(rms::Predict(fit_multiple, conf.int = 0.95))
```



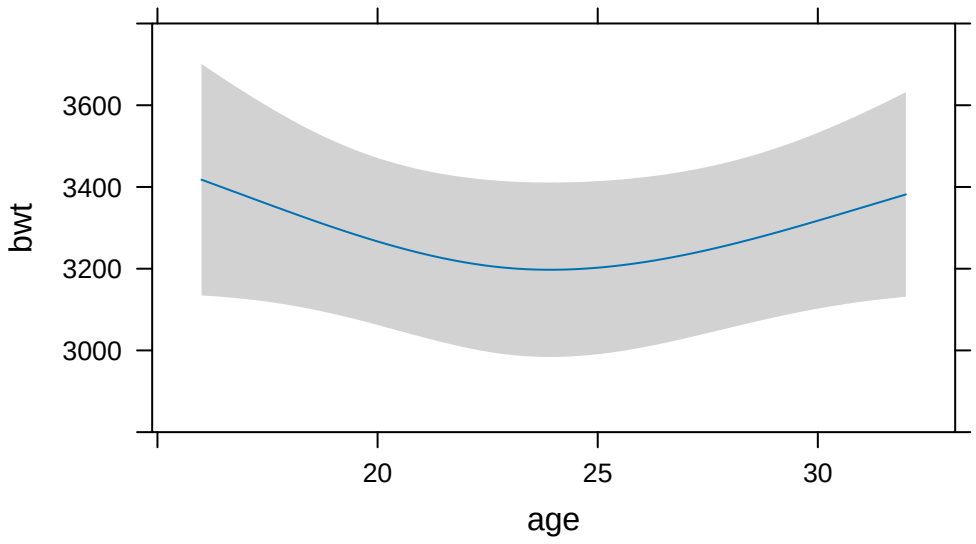
Splines - non-linear affect

```r
fit_spline <- rms::ols(
  bwt ~ rcs(age, 3) + lwt + race + smoke + ptl,
  data = d, x=TRUE, y=TRUE)
anova(fit_spline)
```

```
                Analysis of Variance          Response: bwt

 Factor     d.f. Partial SS MS         F     P
 age           2   1505754    752877.1  1.67 0.1911
  Nonlinear    1   1498328   1498327.8  3.32 0.0699
 lwt           1   1961667   1961666.9  4.35 0.0384
 race          2   6732392   3366196.2  7.47 0.0008
 smoke         1   4552033   4552032.9 10.10 0.0017
 ptl           1   1786855   1786854.9  3.96 0.0480
 REGRESSION    7  18376782   2625254.6  5.82 <.0001
 ERROR       181  81592874    450789.4
```

```r
plot(rms::Predict(fit_spline, age, conf.int= 0.95))
```



Adjusted to:lwt=121 race=White smoke=No ptl=No

diagnosis of patterns - What do we do to ensure data is ok

#diagnostic plots

```
library(rms)

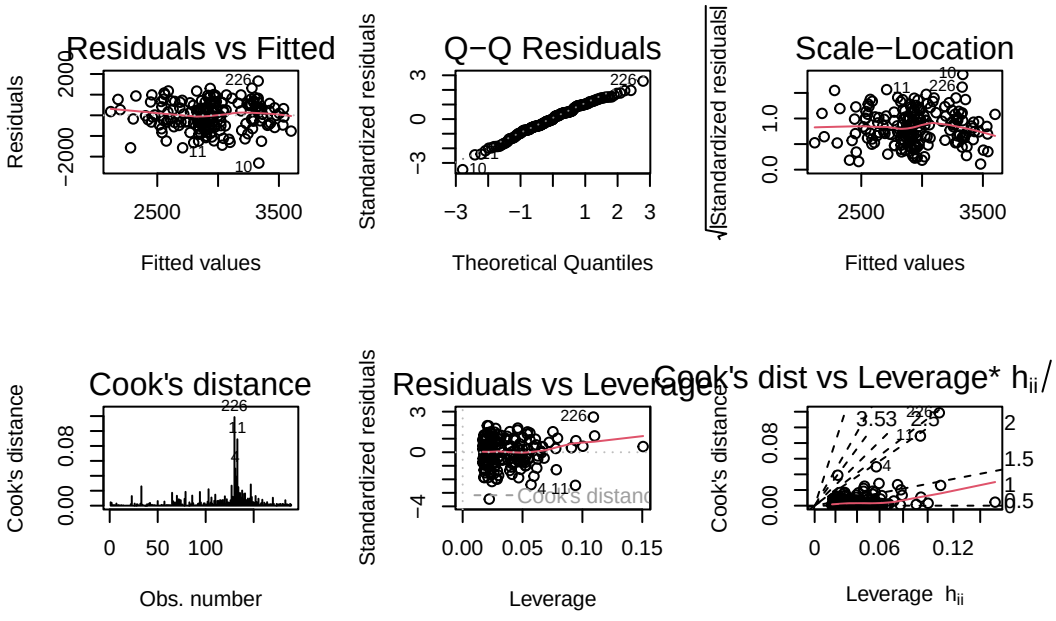
# 1. Fit identical models
formula <- bwt ~ age + lwt + race + smoke + ptl

fit_rms <- ols(formula, data = d, x = TRUE, y = TRUE)
fit_lm  <- lm(formula, data = d)

# 2. Standard regression diagnostics using lm()
par(mfrow = c(2, 3))

#pre-specified which 1-6 for rms function of diagnostics

plot(fit_lm, which = 1)  # Residuals vs fitted: linearity
plot(fit_lm, which = 2)  # Normal Q-Q: residual normality
plot(fit_lm, which = 3)  # Scale-location: constant variance
plot(fit_lm, which = 4)  # Cook's distance: influential observations
plot(fit_lm, which = 5)  # Residuals vs leverage
plot(fit_lm, which = 6)  # Cook's distance vs leverage
```



```r
par(mfrow = c(1, 1))

# 3. Multicollinearity using rms
vif(fit_rms)
```

```
       age        lwt race=Black race=Other  smoke=Yes    ptl=Yes
  1.120393   1.157313   1.196240   1.347178   1.211736   1.083214
```

```r
# 4. ANOVA: predictor contributions
anova(fit_rms)
```

```
                Analysis of Variance          Response: bwt

 Factor     d.f. Partial SS   MS          F     P
 age           1      7426.376    7426.376  0.02 0.8987
 lwt           1   1934011.119 1934011.119  4.24 0.0410
 race          2   6755126.332 3377563.166  7.40 0.0008
 smoke         1   4572735.498 4572735.498 10.02 0.0018
 ptl           1   2053083.370 2053083.370  4.50 0.0353
 REGRESSION    6  16878454.526 2813075.754  6.16 <.0001
 ERROR       182  83091201.284  456545.062
```

```r
# 5. Influential observations using rms
which.influence(fit_rms)
```

```
$Intercept
[1] "226" "4"   "11"  "33"  "57"

$age
[1] "119" "226" "10"  "11"  "22"

$lwt
[1] "202" "226" "15"  "28"

$race
[1] "119" "172" "4"   "10"  "11"  "16"  "18"  "28"

$smoke
[1] "180" "4"   "10"  "11"

$ptl
[1] "154" "161" "188" "199" "223" "4"   "13"  "18"  "29"
```

penalization ridge, shrinkage and lasso net

```r
# Fit the unpenalized model with x and y retained
fit_rms <- rms::ols(
  formula,
  data = d,
  x = TRUE,
  y = TRUE
)

# Evaluate candidate penalty values
ridge_trace <- rms::pentrace(
  fit_rms,
  penalty = c(0, 5, 10, 25, 50)
)

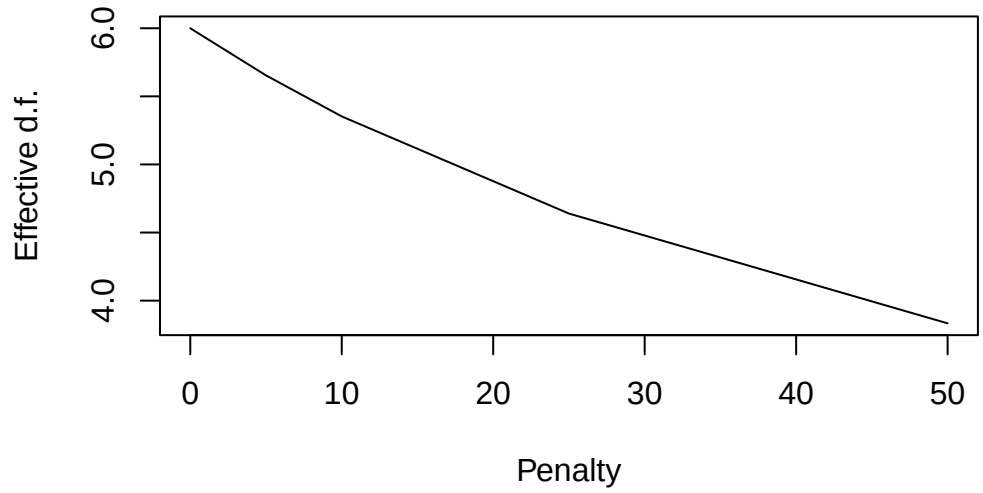
# Display results
print(ridge_trace)
```

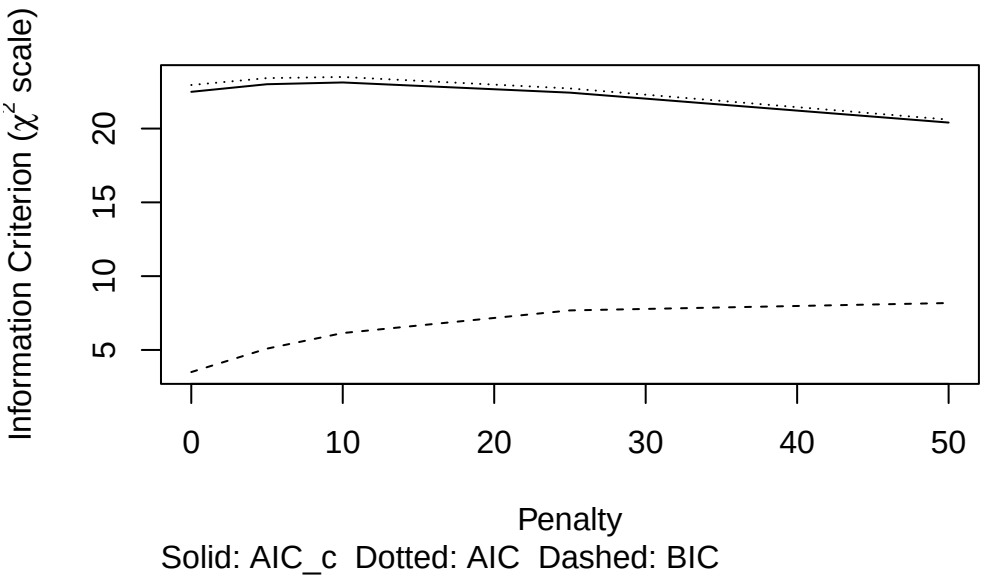
```
Best penalty:

 penalty       df
      10 5.352658

 penalty       df      aic      bic    aic.c
       0 6.000000 22.95137 3.500888 22.48983
       5 5.653881 23.41739 5.088940 23.00477
      10 5.352658 23.49637 6.144406 23.12403
      25 4.639503 22.71951 7.679413 22.43412
      50 3.833382 20.60698 8.180120 20.40576
```

```r
plot(ridge_trace)
```





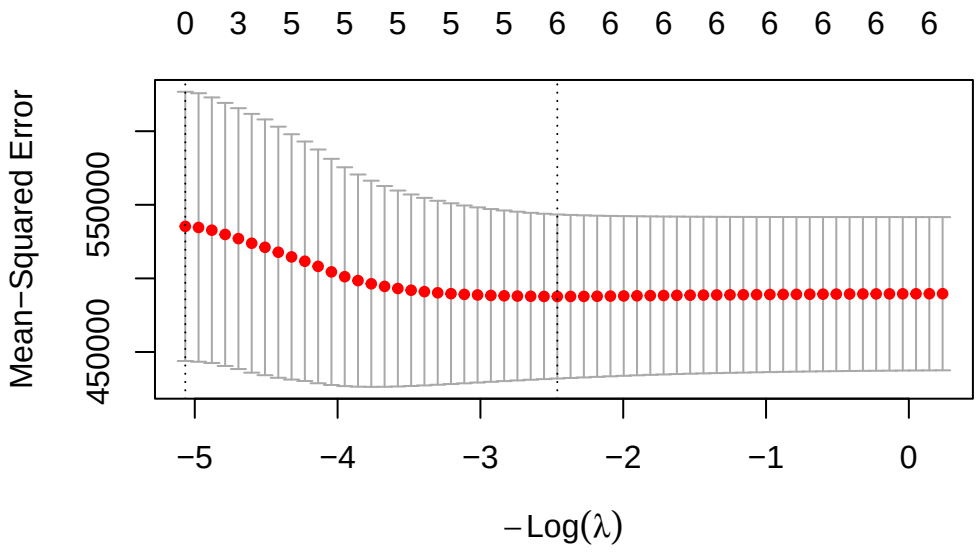
Solid: AIC_c Dotted: AIC Dashed: BIC

```
set.seed(2026)

cv_lasso <- glmnetUtils::cv.glmnet(
  formula,
```

```
  data = d,
  family = "gaussian",       # Correct spelling
  alpha = 1,                 # LASSO regression
  nfolds = 5,                # 5-fold cross-validation
  type.measure = "mse",      # Mean squared error
  standardize = TRUE,
  use.model.frame = TRUE
)

plot(cv_lasso)
```




```
coef(cv_lasso, s = "lambda.min")
```

```
7 x 1 sparse Matrix of class "dgCMatrix"
              lambda.min
(Intercept) 2857.1618010
age            0.2271451
lwt            3.2824683
raceBlack   -430.1574006
raceOther   -328.6767627
smokeYes    -317.8252828
ptlYes      -276.1685191
```

deploys PCR and partial least square